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Hollow Knight - Charms, Notches, Upgrades, Shops, and more

This project will map transactions between useful game items, such as charms, notches, mask shards, and spells. These interact with multiple locations, sub-locations, shops, vendors, side-quests and more.

The diagram is an Entity-Relationship (ER) model for a game system. It includes the following components:

- Entities:**
 - Location:** Attributes include Name (underlined) and id (circled). It is connected to **Charm Notch** via an "is located at" relationship (1:N) and to **Mask Shard** via an "is located at" relationship (1:N).
 - Vendor:** Attribute is Name (underlined). It is connected to **Charm Notch** via an "is sold by" relationship (1:N) and to **Mask Shard** via an "is sold by" relationship (1:N).
 - Charm Notch:** Attribute is id (circled). It is connected to **Location** (1:N), **Vendor** (1:N), and **Requires** (1:N).
 - Mask Shard:** Attribute is id (circled). It is connected to **Location** (1:N), **Vendor** (1:N), and **Requires** (1:N).
 - Condition:** Attributes include text (underlined) and cost (underlined). It is connected to **Requires** (1:M) and **happens in** (1:N).
- Relationships:**
 - is located at:** Connects **Location** and **Charm Notch** (1:N).
 - is sold by:** Connects **Vendor** and **Charm Notch** (1:N).
 - is located at:** Connects **Location** and **Mask Shard** (1:N).
 - is sold by:** Connects **Vendor** and **Mask Shard** (1:N).
 - Requires:** Connects **Charm Notch** (1:N), **Mask Shard** (1:N), and **Condition** (1:M).
 - happens in:** Connects **Condition** (1) and **Location** (N).

DB Schema

an asterisk indicates primary key

Charms Locations

id*	location name
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Charms Requirements

id*	condition text
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Charms Vendor

id*	vendor name	cost
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Mask Shards Locations

id*	location name
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Mask Shards Requirements

id*	condition text
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Mask Shards Vendor

id*	vendor name	cost
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Vendor Locations

vendor name*	location name
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Implementation of Transactions

For this project, I will implement all 4 CRUD actions to simulate Team Cherry adding a DLC that would add/replace/delete old charm notches and mask shards.

DBMS

I will use MySQL, as it is a free DBMS.

Programming Language

I will be using Rust as my programming language of choice.